Model Predictive Control Spring 2026
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Solution 04
Optimization

Exercise 1

Are the following sets convex? Justify your answers.

$P_1 = \{x \in \mathbb{R}^n \mid Ax \leq b\}$
$P_2 = \{x \in \mathbb{R}^n \mid x^\top Hx \leq 2, H \succ 0\}$
$P_3 = \{x \in \mathbb{R}^n \mid x^\top x = 0\}$
$P_4 = \{x \in \mathbb{R}^n \mid x^\top x = 2\}$
Set of non-negative functions: $P_5 = \{f(x) | f(x) \geq 0 \ \forall x \in \mathbb{R}^n\}$
Set of positive semi-definite matrices: $P_6 = \{M \in \mathbb{R}^{n\times n} | M^\top = M,\ x^\top Mx \geq 0 \ \forall x \in \mathbb{R}^n\}$

Solution:

- P_1: Polyhedra are convex sets (Lec. 3, Slide 20)
- P_2: Is a convex set: By the second-order condition for convexity (Lec 3, Slide 28) we have that $f(x) = x^\top Hx$ is a convex function since $\nabla^2 f(x) = H$ is positive definite. We know that if f is convex $\Rightarrow$ sublevel sets of f are convex sets (Lec 3, Slide 29).

  Alternatively: Can be restated as Ellipsoidal set, which is convex (Lec 3, Slide 21)
- P_3: Is a convex set: The set is given by $\{0\}$. By the definition of convex sets $\lambda\cdot 0+(1-\lambda)\cdot 0 = 0 \in P_3$
- P_4: Is not a convex set: Using the definition of a convex set $\mathcal{X}$:
  A set $\mathcal{X}$ is convex $\Leftrightarrow \forall x, y \in \mathcal{X}, \lambda \in [0, 1] : \lambda x + (1 - \lambda)y \in \mathcal{X}$ we falsify convexity with a counter example. Pick $x = \sqrt{2}, y = -\sqrt{2}, \lambda = 0.5$ for which $0.5x + (1 - 0.5)y = 0 \notin P_4$.
- P_5: Is a convex set: $f_1(x) \in P_5, f_2(x) \in P_5 \Rightarrow \underbrace{\lambda}_{\geq 0} \underbrace{f_1(x)}_{\geq 0} + \underbrace{(1-\lambda)}_{\geq 0} \underbrace{f_2(x)}_{\geq 0} =: g(x) \geq 0 \ \forall x \in \mathbb{R}^n \Rightarrow g(x) \in P_5$.
- P_6: Is a convex set: Lets pick two matrices $X \in P_6, Y \in P_6$. Consider $L = \lambda X + (1 - \lambda)Y$ for any $\lambda \in [0, 1]$. By definition of P_6 we have $X^\top = X$ and $Y^\top = Y$ and therefore $L = \lambda X + (1 - \lambda)Y = \lambda X^\top + (1 - \lambda)Y^\top = L^\top$. Also by definition of P_6, $x^\top Lx = \lambda x^\top Xx + (1 - \lambda)x^\top Yx \geq 0$. Hence L is an element of P_6.

Exercise 2

Consider the optimization problem

$$
\begin{aligned}
P: \quad x_p^\star = \arg \min_{x\in\mathbb{R}^2} \ & \|x\|_p \\
\text{s.t. } & x_1 + x_2 = 3 \\
& (x_2 - 2)^2 \leq 1
\end{aligned}
$$

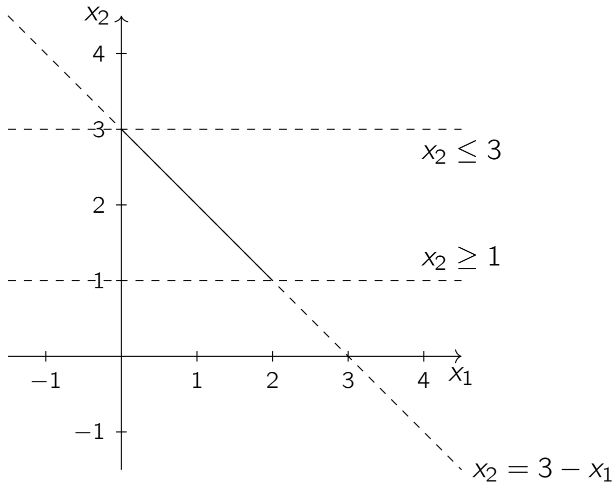
with $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.

i) Draw the feasible set of P.

ii) State the set of all optimal solutions $X_1^\star$ for $p = 1$, $X_2^\star$ for $p = 2$, and $X_\infty^\star$ for $p = \infty$.

Solution:

i) We draw the inequality and equality constraints of P. Note that the resulting feasible set of P does not depend on the objective function of P.



ii)

- 1-norm: $X_1^\star = \{x \mid x_1 + x_2 = 3,\ (x_2 - 2)^2 \leq 1\}$
- ∞-norm: $X_\infty^\star = \{x = \begin{bmatrix} 1.5 \\ 1.5 \end{bmatrix}\}$
- 2-norm: $X_2^\star = \{x = \begin{bmatrix} 1.5 \\ 1.5 \end{bmatrix}\}$

Exercise 3

Show that the duality gap is zero and the KKT conditions are satisfied for the following convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & c^T x \\ \text{subj. to} \quad & x_1 \geq 1 \\ & x_2 \geq 2 \end{aligned}$$

with $c = [3, 4]^T$.

Solution:

By inspection, the primal solution is $x^* = [1, 2]^T$.

The inequalities can be written as $Cx \leq e$

$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} -1 \\ -2 \end{bmatrix}$$

The dual is written as (See Slide 48 Lec.3)

$$\begin{aligned} \max_{\lambda \in \mathbb{R}^2} \quad & -e^T \lambda \\ \text{subj. to} \quad & C^T \lambda + c = 0 \\ & \lambda \geq 0 \end{aligned}$$

where we can find the solution $\lambda^* = [3, 4]^T$.

The duality gap is indeed equal to zero, since the cost function is equal to 11 for both the primal and the dual. We could have guessed it earlier since Slater condition holds (there exists a strictly feasible point). We can also check that all the KKT conditions are satisfied

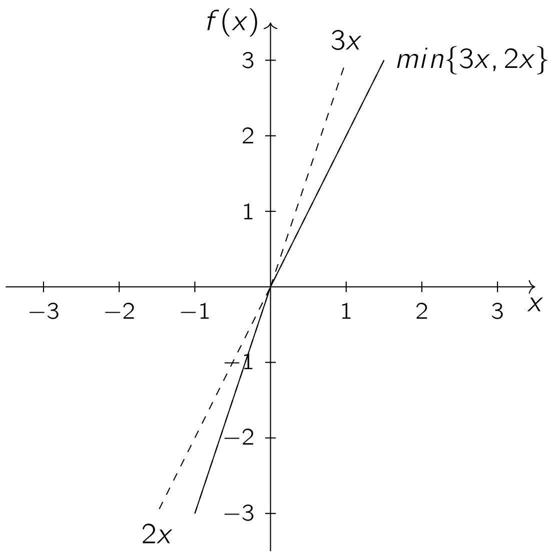
1) $g_i(x_i^*) \leq 0$
2) $\lambda_i^* \geq 0$
3) $\lambda_i^* g_i(x_i^*) = 0$
4) $\nabla_x L(x, \lambda) = c + C^T \lambda^* = 0$

Exercise 4

Is $f(x) = \min\{3x, 2x\}$ a convex function? Justify your answer.

Solution:

Graphical interpretation



Pick $x = 1, y = -1, \lambda = 0.5$:

$$\begin{aligned} f(0) &\leq \lambda f(1) + (1-\lambda) f(-1)? && (1) \\ \min\{0, 0\} &\leq 0.5 \min\{3, 2\} + 0.5 \min\{-3, -2\}? && (2) \\ 0 &\leq 1 - 1.5? && (3) \\ 0 &\leq -0.5 \rightarrow \text{contradiction.} && (4) \end{aligned}$$